KaamPass: Work History for Informal Workers

Problem Statement

Informal workers often rely on verbal agreements, WhatsApp messages, word-of-mouth referrals, and personal connections to find work. Although many workers have years of experience, they may not have a professional profile, verified employment history, or proof of completed jobs. Due to lack of proof when a good opportunity comes, they can easily miss out like if like Saudia have job vacancies open for experienced electricians, an experienced electrician in Pakistan or any other third world country who works informally wouldn't have proof of work or any kind of job completion history to apply for that job. Even though he had experience needed for that job wouldn't be able to qualify for that job.

Employers also struggle to find professional for the wiring, plumbing or paint jobs they have to go out in the market ask for a relevant person for the job. Energy and time is lost in this manual process. Even getting a leak fix can cost hours. Employers also struggle to compare workers, understand their experience, and keep clear records of job agreements.

Project Information

- Project name: KaamPass
- Category: Community and Social Impact / Productivity and Business Solution
- Creator and sole contributor: Abdullah Mehmood
- License: MIT License
- GitHub repository: https://github.com/Abdullahxbytes/KaamPass

KaamPass is currently a functional local prototype. Future work includes deployment, real-user testing, phone-based onboarding, and stronger verification methods for skills and payments.

Relevant SDGs:

SDG 8: Decent Work and Economic Growth

KaamPass helps indivisual informal workers like plumbers, electricians and etc build a professional record of their completed work. Electricians, plumbers, and painters(for now) can show their skills, experience, verified jobs, and feedback to future employers. This can help workers gain trust, find better opportunities, and carry their work history along them.

SDG 10: Reduced Inequalities

Many skilled workers do not have formal employment letters, salary slips, or digital proof of experience. KaamPass gives them a simple way to create a professional identity without requiring formal company employment. The app supports English, Roman Urdu, and Urdu to ensure more people can use it comfortably. This empowers users to leverage modern technologies, showcase their work experience, and access the same opportunities typically available to white-collar professionals

SDG 9: Industry, Innovation and Infrastructure

KaamPass uses a digital workflow to improve how informal work is agreed, recorded, and verified. It combines job requests, bids, agreements, job completion, feedback, and public work records in one platform. This creates a practical digital tool for a community that often depends on informal systems.

Solution Overview

KaamPass is a web platform that helps informal workers create a portable digital work identity. The current prototype focuses on electricians, plumbers, and painters only to deliver the best possible result for now. The system also supports Roman Urdu and Urdu support with right-to-left Urdu layout to make it easier for the workers who are not well learned in reading and writing

Workers can create a profile with their profession, skills, experience, service area, availability, and typical rate. Employers can post jobs with descriptions and optional problem images. Relevant workers can view matching jobs and submit bids with their price, estimated days, and a short message.

The employer selects a bid, which creates an agreement. After the job is completed, the employer records the completed days and amount paid. The worker confirms the information before it becomes a verified work record. This record then contributes to the worker's job history.

Main Features

- Worker and employer accounts with separate permissions.
- Profession specific profiles for electricians, plumbers, and painters.
- Employer job posting with optional problem images.
- Matching jobs based on a worker's saved profession.
- Worker bidding with price, estimated days, and message.
- Employer bid comparison and worker selection.
- Digital agreements that workers can accept or reject.
- Employer job completion and worker confirmation.
- Verified work history, work days, and repeat-employer statistics.
- Employer feedback after a verified job.
- Worker-controlled public profile and QR sharing link.
- Portfolio images with privacy controls.
- English, Roman Urdu, and Urdu support with right-to-left Urdu layout.

Technical Approach

KaamPass uses React, TypeScript, and Vite for the web interface. The backend uses Express and TypeScript. Supabase provides authentication, PostgreSQL database storage, file storage, and SQL migrations.

The application uses role-based access control so workers and employers can only perform actions relevant to their role. Input is validated through Zod schemas, and important workflow actions such as accepting agreements, completing jobs, and confirming work are handled through

controlled server-side processes. The project also includes automated checks using ESLint, Prettier, TypeScript, and Vitest.

Original Contribution and Value

This same problem occurs in multiple third world countries like Bangladesh, Sri Lanka , Nigeria , Nepal and many more. The informal employment system or day jobs in these countries is all have this same problem. Adding a new language to the project can be great contribution and increase the user base massively.

KaamPass does not only provide a job board. Its main contribution is turning informal work into a worker-owned, verified history. The worker does not depend only on a current employer or informal references because completed work can become part of a portable record.

The mutual confirmation process is important because an employer cannot create a verified history record alone. The worker must confirm the completed work and payment details. This creates a clearer and more trustworthy process for both sides.

Demonstration Flow

1. An employer posts a job request.
2. A matching worker views the request and submits a bid.
3. The employer compares bids and selects a worker.
4. The worker accepts the generated agreement.
5. The employer marks the work as completed.
6. The worker confirms the work and payment.
7. The verified job appears in the worker’s work history.
8. The worker can choose to share a public profile through a QR link.